Galois and Quintic Formula

Patrick Was

1 Definitions

1.1 Groups

First, some notation. In order to make this more readable, define the symbol "*" to mean some binary operation (meaning it is performed on two elements/numbers), such as addition, multiplication, subtraction, or division. Also, let G be some set. A **group** is a set of elements that satisfies the following properties:

- **Closure.** $\forall a, b \in G, a * b \in G$
  For example, the set $\{1, 2\}$ is closed under multiplication, since $1 \cdot 2 = 2$ and $2 \cdot 1 = 2$, which are both in the set. However, it is not closed under addition, since $1 + 2 = 3$, which is not in the set.

- **Identity.** $\exists e \in G \mid \forall x \in G, e * x = x * e = x$
  The set $\{0, 5\}$ is an obvious example of a set closed under addition. It is clear that $5 + 0 = 0 + 5 = 5$, and also that $0 + 0 = 0 + 0 = 0$. So, both elements have an identity element, that being 0.

- **Inverse.** $\forall x \in G, \exists y \in G \mid y * x = x * y = e$
  This e is the same as from before, the identity element. The set $\{-1, 1\}$ may appear to satisfy this property, since generally -1 is the inverse of 1, $-1 + 1 = 0$, and since there is no identity element 0 in the set, this set does not follow this property. $\{-1, 0, 1\}$, however, does.

- **Associativity.** $\forall x, y, z \in G, (x * y) * z = x * (y * z)$
  This is analogous to associativity under the set of real numbers.
  It is not the most well-defined to attempt associativity on an set with two elements, such as $\{-1, 1\}$. However, we do know that $-1, 1 \in \mathbb{R}$, and all elements of the real numbers satisfy associativity under multiplication, so this set must satisfy associativity under multiplication. One of the elements can also be used twice in the demonstration.

An **abelian group** is one in which $\forall x, y \in G, x * y = y * x$. Basically, it is analogous to the commutative property when dealing with numbers. Any set with complex numbers will satisfy this property, since there it would just be the commutative property, which is a property that extends to that set.

A **cyclic group** is one in which $G = \{g^n \mid n \in \mathbb{Z}\}$ or $G = \{ng \mid n \in \mathbb{Z}\}$. Basically, this just means that every element of G can be formed by a single **generator element** through either multiplying by some constant or raising it to a power; in other words, it acts on itself somehow to form the other elements. G is typically defined more rigorously as a cyclic group in terms of the **subgroup** of the element generator, $\langle g \rangle$, if $\langle g \rangle = \{g^n \mid n \in \mathbb{Z}\} = G$, or if the group is abelian and additive, then $\langle g \rangle = \{ng \mid n \in \mathbb{Z}\} = G$. This group is also typically written as C_n for its n elements.

An example of a cyclic group is $(\mathbb{Z}, +)$, which represents the additive group of integers. This is because every element can be generated using 1 and -1 simply by multiplying by some constant n. $(\mathbb{Z}_{11}^{+}, \times_{11})$, the set of positive integers mod 11 under mod 11 multiplication, is also a cyclic group with one of its group generators being 6. As n goes from 1-10, 6^n hits every element of the group: $\{6, 3, 7, 9, 10, 5, 8, 4, 2, 1\}$, in that order.

A **symmetric group** is a group of all permutations on a set, with the binary operation being the composition of functions. If the set contains n elements, we know that the group will contain $n!$ elements, since that encompasses all permutations of those elements. The idea of permutations will be expanded upon later. The symmetric group is also typically written as S_n for its n elements.

An **alternating group** is the set of all **even** permutations in S_n. In order to define what it means for a permutation to be even, it's important to know that a **transposition** is the permutation of 2 elements of a set. Therefore, every 2-cycle is a transposition. A permutation is even if one if its transposition representations consists of an even number of transpositions. In other words, if there was a sequence of transpositions that led to the identity element of the set, there would need to be an even number of transpositions in that sequence. This will be expanded upon later. This group is typically written as A_n for its n elements.

1.2 Fields

A **field** is a set F which is closed under the two binary operations of addition and multiplication such that

- F is an abelian group under addition.
- $F \backslash \{0\}$ is an abelian group under multiplication.

A field K is said to be a **field extension** of a field F if F is a subfield of K. This is denoted as K/F, though field extensions are more generally written in terms of one number at a time, written as $F(\alpha)$.

For example, consider the set of rational number, $\mathbb{Q}$. If we were trying to solve for all solutions to the polynomial $x^2 - 3$, we would not be able to under the field of $\mathbb{Q}$. We do know that the roots are $\pm\sqrt{3}$, so if we were to extend our field to include $\sqrt{3}$, we would be able to find the solutions. This could be written as $\mathbb{Q}(\sqrt{3})$, which is essentially equivalent to $\{a + b\sqrt{3} \mid a, b \in \mathbb{Q}\}$. Any algebraic combination involving rational numbers will therefore be in the field $\mathbb{Q}(\sqrt{3})$. The expression $x^2 - 3$ will therefore **split**, or factor, in $\mathbb{Q}(\sqrt{3})[x]$, which is the field of the coefficients of this particular quadratic. The **splitting field** is therefore the smallest field containing the solutions to the polynomial.

As a further example, we could express $\mathbb{Q}(\sqrt{3}, \sqrt{5}) = \{a + b\sqrt{3} + c\sqrt{5} + d\sqrt{15} \mid a, b, c, d \in \mathbb{Q}\}$, but we cannot have an element like $e\sqrt{10}$, since this would involve multiplying $\sqrt{5}$ by $\sqrt{2}$, which is not an element of our field.

A cool fun fact is that $\mathbb{R}(i)$ is equivalent to $\mathbb{C}$.

An **isomorphism** is a special mapping between two groups that sets up a one-to-one correspondence with the group that respects the binary operations pertaining to each group. A more rigorous definition is the following:

A group isomorphism from $(G, *)$ to (H, Δ) is a bijection from G to H such that for all x and y in G:

$$f(x * y) = f(x)\Delta f(y)$$

An **automorphism** is very similar, except that it is a mapping to itself with the same test but the two operations must be the same. Basically, it is an isomorphism from the group to itself.

1.3 Permutations

A **permutation** of a set S is a bijective function such that $S \to S$.

1.3.1 Cycle Notation

When permuting a set, **cycle notation** is typically used. It is a simple way to express what each element of the original set permutes to.

Consider the set $\{1, 2, 3, 4, 5\}$. If we wanted to compute every element to its following element, we could write it as the cycle (1, 2, 3, 4, 5). This new set would look like $\{2, 3, 4, 5, 1\}$. We have 1 permute to 2, 2 permute to 3, and so on until 5 permutes to the beginning of the set, 1.

Using the same set, we could write the permutation that permutes all of the sets' elements to itself. This would be (1)(2)(3)(4)(5). The parentheses denote the end of the cycle, so we see the set remains identical. Similarly, in (1, 2)(4, 5), we have 1 permuting to 2 and 2 permuting to 1 as well as 4 permuting to 5 and 5 permuting to 4, but the remaining elements permute to themselves. This kind of permutation that contains **disjoint cycles**, since they do not affect each other. In this case, it is the product of two disjoint cycles.

1.3.2 Functional Cycle Notation

Since we know every permutation is a function, our symmetric group is a group composed of these functions. Since every element is a function, we can see what happens if we compose these functions with each other. In tune with how compositions of functions are typically denoted, $f \circ g$ will be equivalent to fg.

If we consider the two permutations from earlier, (1, 2, 3, 4, 5) and (1, 2)(4, 5), we can denote each as σ and τ respectively. We could find $\sigma\tau$: (1, 3, 4, 5). In this case, 2 actually goes back to map to itself. It's also important to note that this group is not abelian, meaning that the elements will not commute. As an example, $\tau\sigma = (2, 3, 5)$, meaning 1 maps to itself and 4 maps to itself. We see that $\sigma\tau \neq \tau\sigma$.

We can also compose a function on itself. For example, $\sigma^2 = \sigma \circ \sigma$.

We also know that if permutations $\alpha\beta$ are two disjoint cycles, then $\alpha\beta$ do indeed commute. We can see that this is true by a quick example: since $(1, 2)$ and $(3, 4)$ do not affect each other in any way, then $(1,2)(3,4) = (3,4)(1,2)$.

In addition, we know that since this is a group, there must be an identity element function. This would just be $(1)(2)(3)(4)(5)$, since every element is permuted to itself.

There must also be a permutation σ^{-1} for any element σ such that $\sigma^{-1}\sigma$ or $\sigma\sigma^{-1}$ equals to the identity element.

1.3.3 More on Alternating Groups

A **simple** alternating group is one that only consists of its entire group (the improper subgroup) and its trivial group of order one, or of one element as its **normal subgroups** (which are subgroups that do not change by any sort of conjugation; a subgroup H is normal iff $gHg^{-1} = H$).

There is a very special alternating group that is the foundation of Galois theory: A_5. It is the smallest non-abelian simple group, which eventually paves the way for why there is no general quintic formula solvable by any binary operation.

1.4 Galois groups

A **galois group** is the set of automorphisms of L/K, denoted as either $\mathrm{Gal}(L/K)$ or $\mathrm{Aut}(L/K)$, for automorphism.

2 Quintic formula

Galois's theorem states that an algebraic equation is **solvable** iff its galois group is solvable (such that each of its normal subgroups are abelian).

Though rather difficult, it can be proved that A_5 is the smallest non-abelian simple group, and will therefore make its parent group, S_5, not solvable either. Then, all that needs to be done is to form a connection between the quintic and the group S_5.

The symmetric group S_5 does have some special isomorphisms, though, which means that it will share properties with that group. For instance, if we let $s_1 s_2 s_3, \ldots, s_n$ be the elementary symmetric functions and let $x_1, x_2, x_3, \ldots, x_n$ represent these symmetric functions, then $F(y_1, y_2, \ldots, y_n)$ is a field extension of degree $n!$ of $F(s_1, s_2, \ldots, s_n)$. This makes it and its galois group isomorphic to S_n, since they share the same number of elements.

If we let the group $E = \mathbb{Q}(y_1, y_2, y_3, y_4, y_5)$ and let

$$f(x) = \prod_{i=1}^{5}(x - y_i) = x^5 - s_1x^4 + s_2x^3 - s_3x^2 + s_4x - s_5$$

where s_n is the roots taken n at a time, we know that $s_1, s_2, s_3, s_4, s_5 \in E$, so $f(x) \in E[x]$. If we let $F = \mathbb{Q}(s_1, s_2, s_3, s_4, s_5)$, we see that $f(x) \in F[x]$ and E is the splitting field over F of $f(x)$, since it contains all its roots. The galois group $G(E/F)$ is isomorphic to S_5, and since that group is not solvable, neither is this one, making $f(x)$ not solvable.

The basic premise behind this is that if the roots aren't re-arrangeable if not given their specific values, then it cannot be possible to find a way to manipulate the roots such that they satisfy an algebraic equation. It would likely take an infinite number of nested roots to actually satisfy some polynomial.

3 Other Applications

There are a considerable number of geometric construction problems that are shown to be impossible by Galois theory, since these involve simple binary arithmetic operations. Some such examples include the impossibility of trisecting general angles or doubling the volume of cubes, as well as other n-dimensional cubes.

It is also very useful in RSA encryption, seeing as it leads the way for commutators and other properties in groups that make certain values hard to decode.